

Gesture-Controlled Contactless Switch For Smart Home

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To avoid the risk of contracting Covid-19, it is important not to touch surfaces including switches, doorknobs, and keys that have been frequently used by other people. This project is for a contactless switch that works with hand gestures.

The smart switch includes a sensor that can detect hand movements and translate them into commands for controlling lights, fans, and various other home appliances. The author's prototype for this switch, with its OLED display installed on the wall switchboard, is shown in Fig. 1. The list of components required for designing the gesture-controlled switch is given in Table 1.

Coding

First, install the required libraries in the Arduino IDE. Go to Select → Library Manager → Search APDS9960 and U8g2lib as shown in Fig. 2. Then add each library to Arduino code as shown in Fig. 3. Define Arduino pin numbers for controlling the light and fan. Next, create a setup function where you can initialise the sensor and set the pin mode output for light and fan control as shown in Fig. 4.

Create a loop function to update the sensor readings and use your hand gestures as inputs to control Arduino output pins. Also create a switch condition in the code, as shown in Fig. 5, for specifying the commands based on the type of hand gestures (up, down, left, and right). The icons (arrow symbols) below indicate the gestures and commands to be displayed on the OLED:

- Up: Light on
- Down: Light off
- Left: Fan on
- Right: Fan off

After coding is done, compile it in Arduino IDE and upload the code to Arduino board.

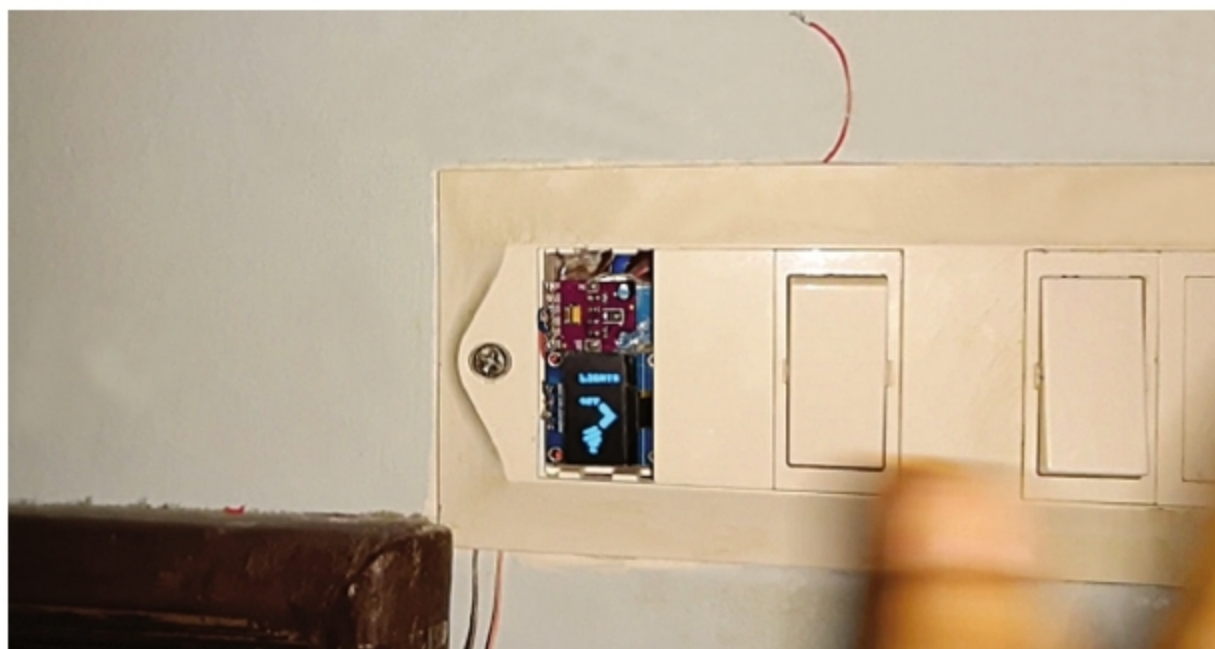


Fig. 1: Prototype of gesture-controlled contactless switch with OLED display

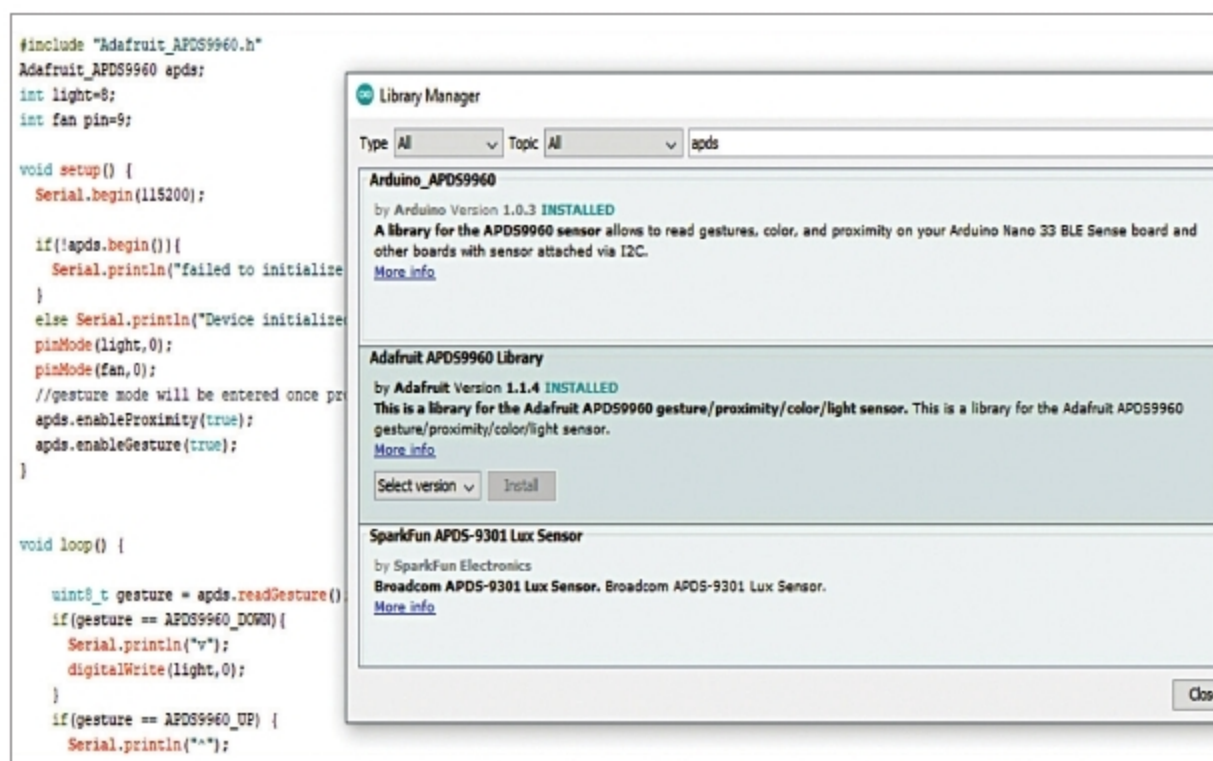


Fig. 2: Library Manager window

Components for Smart Switch			Table 1
Components	Numbers	Description	
Arduino Mini Pro	1	For programming	
SSD1306 OLED	1	OLED display	
2-Channel relay	1	5V DPDT relay module	
5V adaptor	1	For Arduino, relay and APDS	
APDS-9960	1	Gesture sensor module	
Bulb	1	Any light bulb operated through 230 AC mains	
Fan	1	Ceiling fan	